



ShopBuddy: Agentic Shopper Assistant

What if your shopping assistant actually knew the products - and could prove it??

Garima Narang

Gurwinder Singh

Jayram Karan

Pavankumar Naik

Instructor: Prof.
Mohammad Fawaz

RESEARCH PROBLEM & OBJECTIVE

- LLMs produce **confident but hallucinated** product facts - no real-data connection
- Assistants **lose context** on follow-up questions.
- No audit trail**- users can't verify where answers come from
- E-commerce decisions require **trustworthy, grounded** product intelligence

Objective: Build a grounded, agentic, production-deployed e-commerce assistant that retrieves *real evidence before every answer*- with full transparency on each response.

RESEARCH QUESTIONS & HYPOTHESIS

Does a retrieval-grounded, MCP-orchestrated assistant produce more accurate, consistent answers than direct LLM generation?

H1 - Accuracy

RAG + MCP produces higher factual accuracy and lower hallucination vs. direct generation baseline.

H2 - Consistency

MCP standardized context improves **response consistency & relevancy** vs. Agentic RAG w/o MCP

DATA DESCRIPTION & PREPROCESSING

- Scrape**
Flipkart HTML
- Clean**
3-stage cleanup (scrape → ETL → retrieval)
- Ingest**
Schema Validation · LangChain Documents
- Embed**
Gemini embed-001 · 3,072- dim vectors · AstraDB cosine-similarity

236
Vectors in AstraDB

2,500+
Review Chunks

80+
Products Scrapped

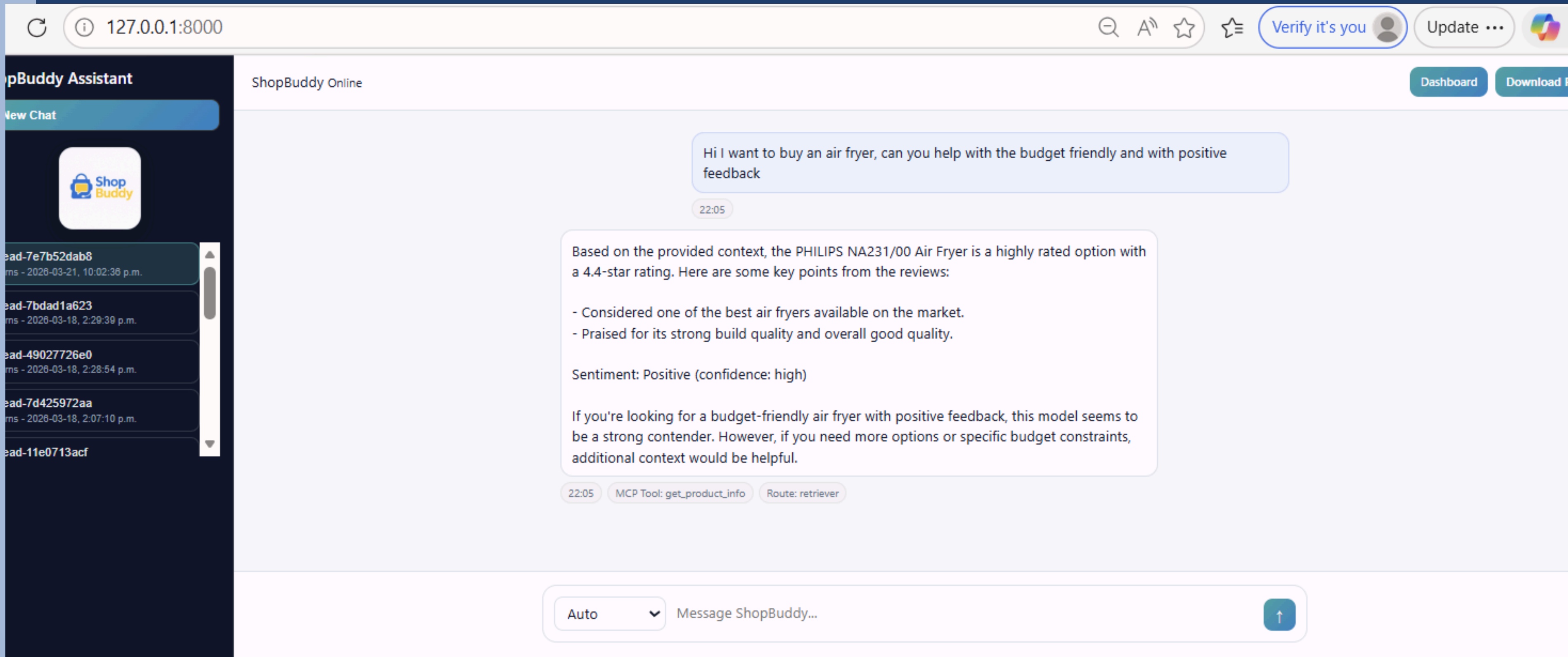
Categories: Phones Laptops Headphones TVs & Tablets

SCAN ME!

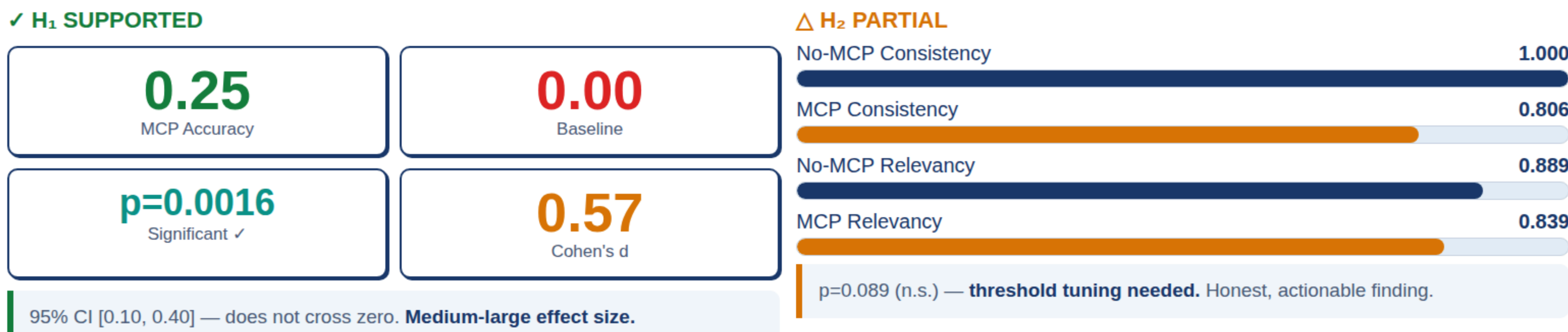


SEE IT IN ACTION

Every answer is grounded in **real product data**, and tells you *exactly where it came from*



KEY FINDINGS & STATISTICAL RESULTS



STARTEGIC & PRACTICAL IMPLICATIONS

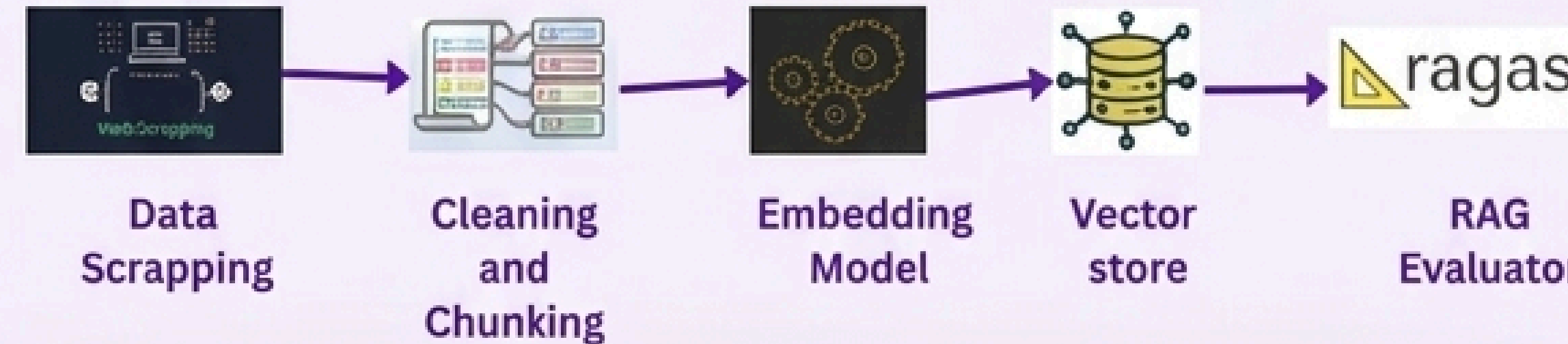
E-Commerce Teams Deploy RAG + MCP for product Q&A — near-zero hallucination on specific product facts.	AI Engineers MCP makes tool backends swappable without workflow rewrites — adopt early for maintainability.
Data & Analytics Route + RAGAS scores are real-time corpus health signals — monitor continuously.	DevOps / MLOps 100% of outages were config errors. DevOps discipline = equally important as model quality.

LIMITATIONS & FUTURE RESEARCH

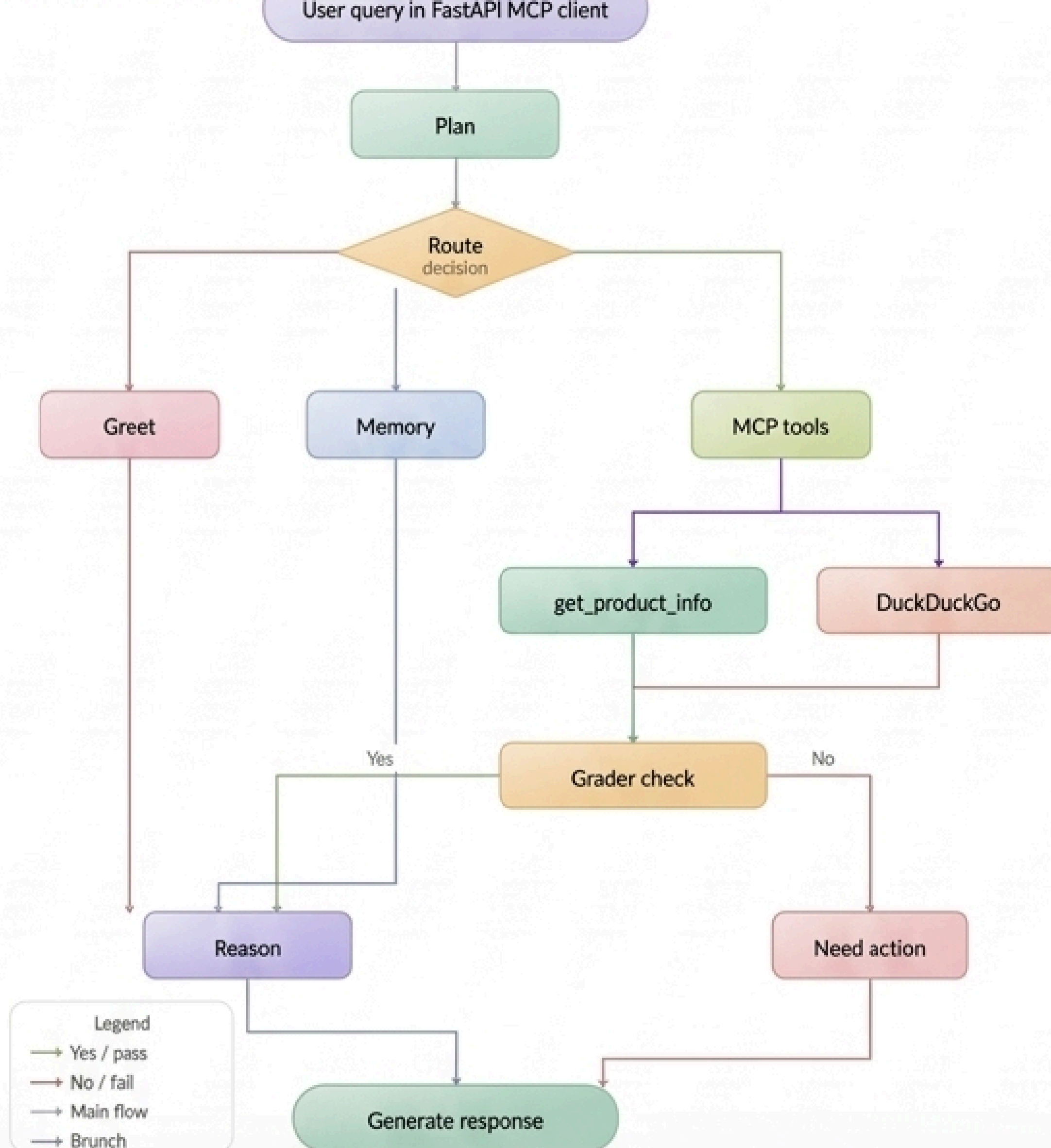
Limitations Small Benchmark n=12, 36 obs — limited power for small effects Single Source Flipkart only — India market bias H₂ Not Supported Threshold 0.3 needs calibration (p=0.089) No User Testing Automated evaluation — no human judges	Future Research Expand Benchmark 50+ queries, more product categories Multi-Platform Amazon, Best Buy — reduce market bias Threshold Tuning Ablation study on RAGAS & MMR lambda_mult Production Scale PostgreSQL · OIDC auth · latency observability
--	--

ARCHITECTURE

DATA LAYER



AGENT LAYER



OPS LAYER

